MD. JALAL ABEDIN



CSE Graduate

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Career Objective: Computer Science and Engineering graduate with an IEEE-published paper and hands-on experience building and deploying software and AI systems (Flask APIs, Streamlit dashboards, HuggingFace Spaces). Seeking an entry-level role in software, data, or AI where I can apply my technical skills, keep learning, and contribute to real-world projects.

Education

North South University — B.Sc. in Computer Science & Engineering — Dhaka, Bangladesh
Relevant: Machine Learning, Neural Networks, Data Science, Computer Vision — 2022-2026
CGPA: **3.00** / **4.00**

Govt. Debendra College — Higher Secondary Certificate (HSC) — Manikganj, Bangladesh
GPA: **4.58** / **5.00** — 2020

Manikgonj Govt. High School — Secondary School Certificate (SSC) — Manikganj, Bangladesh
GPA: **4.89** / **5.00** — 2018

Technical Skills

Programming Languages	Python, C, C++, Java, PHP, SQL
ML & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, LSTM, Bi-LSTM, Transformer, SHAP, RandomizedSearchCV, GridSearchCV
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, EDA, SMOTE, Class Imbalance Handling
AI & Generative AI	Stable Diffusion, LLMs (Qwen2.5-1.5B, LLaMA), VLMs (Qwen2-VL-7B), Computer Vision, Grad-CAM, CLIP
Backend & Deployment	Flask, REST APIs, Streamlit, Gradio, Joblib, Git, GitHub, Google Colab, Jupyter Notebook, Kaggle, VS Code
Spoken Languages	Bangla (Native), English (Fluent)

Research & Applied Projects

XRayMind – AI Radiology Assistant | *PyTorch · DenseNet121 · Grad-CAM · LLaMA · Gradio* — 2026

- **14-disease chest X-ray detector:** trained DenseNet121 on 89,696 NIH ChestX-ray14 images, reaching mean AUC **0.7765**; Grad-CAM heatmaps visualize model focus regions.
- **Deployment:** LLaMA (via Groq API) auto-generates structured radiology reports; full-stack deployment on HuggingFace Spaces (Gradio) + GitHub Pages frontend.

ML Framework for Predicting Thyroid Cancer Recurrence | *Scikit-learn · SMOTE · SHAP · Python* — 2025

- Built an end-to-end pipeline on 5,351 patient records using pre-operative features only, with strict leakage prevention and SMOTE to handle 2.7% class imbalance.
- **97.5% accuracy, AUC = 0.813** — used SHAP for clinical explainability; deployed the final model as a Streamlit web app.

Stock Market Price Predictor | *Keras · TensorFlow · Streamlit · yfinance* — 2024

- Compared LSTM, Bi-LSTM with custom Attention, and Transformer architectures for stock price prediction; deployed as a live dashboard with real-time Yahoo Finance data.
- **Bi-LSTM + Attention achieved best performance:** RMSE = 4.13, R^2 = 0.9556 on out-of-sample data.

Publications

Real-Synthetic Artwork Classification with Deep Transfer Learning: Benchmarking and Web Deployment IEEE ICIICE 2026
Kazi Hasnaine, Asaduzzaman Sunny, Al Amin, **Md. Jalal Abedin**, Ahmed Dhrubo, Mohammad Qayum

- Published at *IEEE ICIICE 2026*, Dubai, UAE (Oral Presentation). Benchmarked DaViT, MViT, GhostNet, RepViT for AI-generated artwork detection — **87.55% accuracy**, **F1 = 0.875**. DOI: 10.1109/ICIICE69672.2026.11565256

References

Dr. Md Adnan Arefeen | *FYP Supervisor — Adversarial ML Research*
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